



(Approved by AICTE, New Delhi & Affiliated to Andhra University)

Pinagadi (Village), Pendurthi (Mandal), Visakhapatnam – 531173



SHORT-TERM INTERNSHIP

By

Council for Skills and Competencies (CSC India)

In association with

ANDHRA PRADESH STATE COUNCIL OF HIGHER EDUCATION

(A STATUTORY BODY OF THE GOVERNMENT OF ANDHRA PRADESH) (2026–
2027)

PROGRAM BOOK FOR
SHORT-TERM INTERNSHIP

Name of the Student: Ms.GUDALA SWATHI

Registration Number: 323129510017

Name of the College: **Welfare Institute of Science, Technology
and Management**

Period of Internship: From: 21-05-2026 To: 21-07-2026

Name & Address of the Internship Host Organization

Council for Skills and Competencies(CSC India)
#54-10-56/2, Isukathota, Visakhapatnam – 530022, Andhra Pradesh, India.

Andhra University

2026

An Internship Report on

STUDENT EXAMINATION SCORE PREDICTION USING ACADEMIC HISTORY AND STUDY HABIT ANALYTICS

Submitted in accordance with the requirement for the degree of

Bachelor of Technology

Under the Faculty Guideship of

Mrs.G.M.Sowjanya

Department of CSE

Welfare Institute of Science, Technology and Management

Submitted by:

GUDALA SWATHI

Reg.No: 323129510017

Department of CSE

Department of COMPUTER SCIENCE ENGINEERING

Welfare Institute of Science, Technology and Management

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2026-2027

Instructions to Students

Please read the detailed Guidelines on Internship hosted on the website of AP State Council of Higher Education <https://apsche.ap.gov.in>

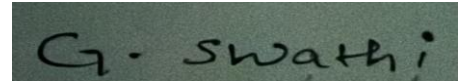
1. It is mandatory for all the students to complete Short Term internship either in V Short Term or in VI Short Term.
2. Every student should identify the organization for internship in consultation with the College Principal/the authorized person nominated by the Principal.
3. Report to the intern organization as per the schedule given by the College. You must make your own arrangements for transportation to reach the organization.
4. You should maintain punctuality in attending the internship. Daily attendance is compulsory.
5. You are expected to learn about the organization, policies, procedures, and processes by interacting with the people working in the organization and by consulting the supervisor attached to the interns.
6. While you are attending the internship, follow the rules and regulations of the intern organization.
7. While in the intern organization, always wear your College Identity Card.
8. If your College has a prescribed dress as uniform, wear the uniform daily, as you attend to your assigned duties.
9. You will be assigned a Faculty Guide from your College. He/She will be creating a WhatsApp group with your fellow interns. Post your daily activity done and/or any difficulty you encounter during the internship.
10. Identify five or more learning objectives in consultation with your Faculty Guide. These learning objectives can address:
 - a. Data and information you are expected to collect about the organization and/or industry.
 - b. Job skills you are expected to acquire.
 - c. Development of professional competencies that lead to future career success.
11. Practice professional communication skills with team members, co-interns, and your supervisor. This includes expressing thoughts and ideas effectively through oral, written, and non-verbal communication, and utilizing listening skills.
12. Be aware of the communication culture in your work environment. Follow up and communicate regularly with your supervisor to provide updates on your progress with work assignments.

Instructions to Students (contd.)

13. Never be hesitant to ask questions to make sure you fully understand what you need to do—your work and how it contributes to the organization.
14. Be regular in filling up your Program Book. It shall be filled up in your own handwriting. Add additional sheets wherever necessary.
15. At the end of internship, you shall be evaluated by your Supervisor of the intern organization.
16. There shall also be evaluation at the end of the internship by the Faculty Guide and the Principal.
17. Do not meddle with the instruments/equipment you work with.
18. Ensure that you do not cause any disturbance to the regular activities of the intern organization.
19. Be cordial but not too intimate with the employees of the intern organization and your fellow interns.
20. You should understand that during the internship program, you are the ambassador of your College, and your behavior during the internship program is of utmost importance.
21. If you are involved in any discipline related issues, you will be withdrawn from the internship program immediately and disciplinary action shall be initiated.
22. Do not forget to keep up your family pride and prestige of your college.

Student's Declaration

I, **GUDALA SWATHI**, a student of **Bachelor of Technology** Program, Reg. No. **323129510017** of the Department of **Computer Science Engineering** do hereby declare that I have completed the mandatory internship from **21-05-2026** to **21-07-2026** at **Council for Skills and Competencies (CSC India)** under the Faculty Guideship of **Mrs.G.M.Sowjanya** Department of **Computer Science Engineering, Wellfare Institute of Science, Technology and Management.**

A rectangular box containing a handwritten signature in black ink. The signature appears to be 'G. Swathi'.

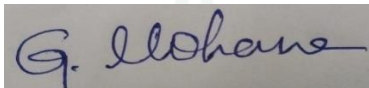
(Signature and Date)

Official Certification

This is to certify that **Ms. GUDALA SWATHI**, Reg.No. **323129510017** has completed his/her Internship at the Council for Skills and Competencies (CSC India) on **STUDENT EXAMINATION SCORE PREDICTION USING ACADEMIC HISTORY AND STUDY HABIT ANALYTICS** under my supervision as a part of partial fulfillment of the requirement for the Degree of **Bachelor of Technology** in the Department of **Computer Science Engineering** at **Welfare Institute of Science, Technology and Management**.

This is accepted for evaluation.

Endorsements



Faculty Guide



Head of the Department
Head Dept of CSE
WISTM Engg. College
Pinagadi, VSP



Principal

Certificate from Intern Organization

This is to certify that **Ms. GUDALA SWATHI** , Reg.No. **323129510017** of **Welfare Institute of Science, Technology and Management**, underwent internship in **STUDENT EXAMINATION SCORE PREDICTION USING ACADEMIC HISTORY AND STUDY HABIT ANALYTICS** And Sentiment Analysis For Social Media Applications at the **Council for Skills and Competencies (CSC India)** from **21-05-2026 to 21-07-2026**.

The overall performance of the intern during his/her internship is found to be **Satisfactory** (Satisfactory/~~Not Satisfactory~~).



Authorized Signatory with Date and Seal

Acknowledgement

I express my sincere thanks to **Dr. A. Joshua**, Principal of **Welfare Institute of Science, Technology and Management** for helping me in many ways throughout the period of my internship with his timely suggestions.

I sincerely owe my respect and gratitude to **Mrs. Nalla Suneetha**, Head of the Department of **Computer Science Engineering**, for his continuous and patient encouragement throughout my internship, which helped me complete this study successfully.

I express my sincere and heartfelt thanks to my faculty guide **Mrs.G.M.Sowjanya** Assistant Professor of the Department of **Computer Science Engineering** for his encouragement and valuable support in bringing the present shape of my work.

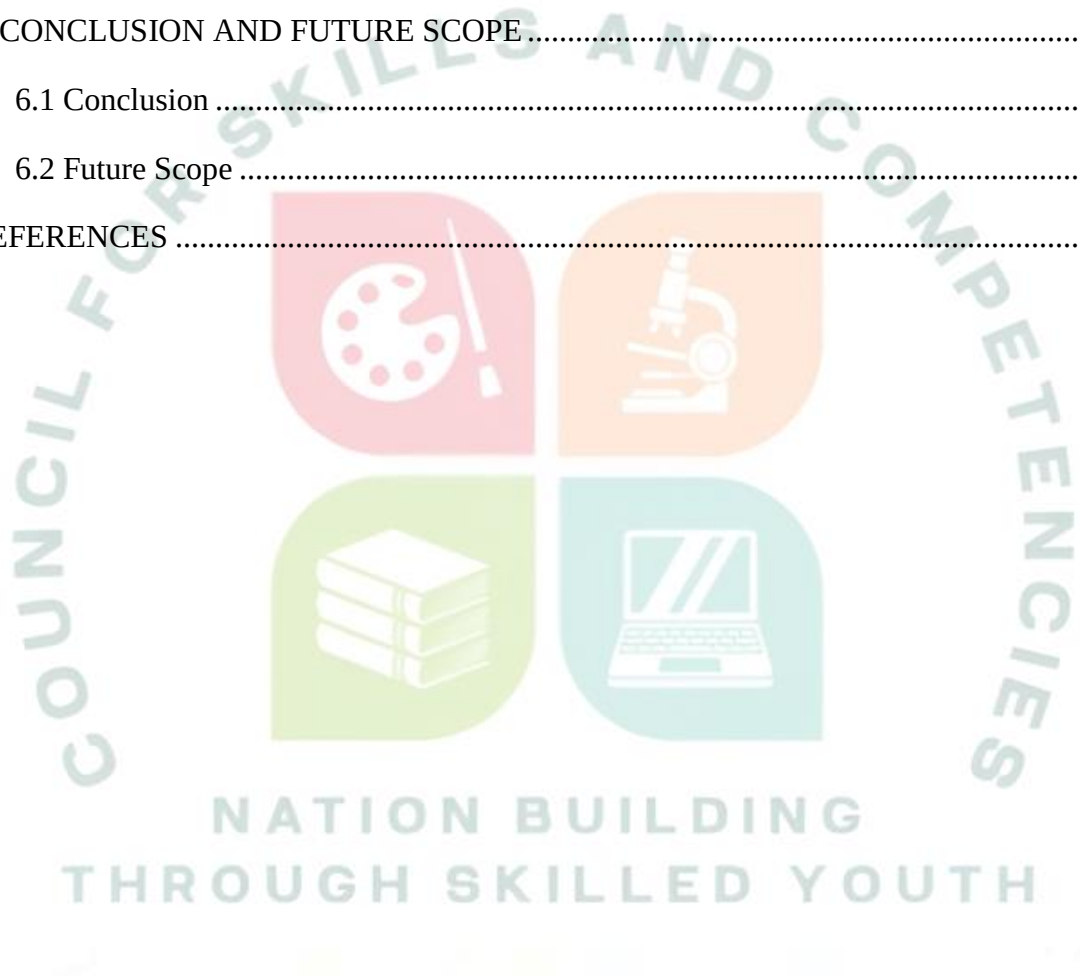
I express my special thanks to my organization guide **Mr. Y. Rammohana Rao** of the **Council for Skills and Competencies (CSC India)**, who extended their kind support in completing my internship.

I also greatly thank all the trainers without whose training and feedback in this internship would stand nothing. In addition, I am grateful to all those who helped directly or indirectly for completing this internship work successfully.

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CHAPTER 1

EXECUTIVE SUMMARY

This internship report provides a comprehensive overview of my internship focused on developing a Student Examination Score Prediction System using Machine Learning and Academic Analytics. The internship spanned an 8-week period and was undertaken to apply predictive analytics to educational challenges. The primary objective of this internship was to gain proficiency in machine learning regression, educational data mining, and predictive modeling to enhance employability skills while solving a critical problem for academic institutions.

1.1 Learning Objectives

During my internship, I learned and practiced the following:

- To design and implement a machine learning predictive engine using Python and Scikit-learn that can accurately forecast student examination scores based on academic history and behavioral metrics.
- To integrate educational data mining techniques for extracting meaningful insights regarding study habits, attendance patterns, and classroom engagement from raw academic logs.
- To implement interactive data visualizations that help faculty members understand the complex relationships between student behavior (study hours, resource usage) and academic outcomes.
- To evaluate multiple regression algorithms including Linear Regression, Random Forest, and Gradient Boosting to determine the optimal model for score prediction.
- To create a scalable analytical framework that provides actionable intelligence for identifying at-risk students, enabling early intervention, and improving overall learning outcomes.

1.2 Outcomes Achieved

Key outcomes from my internship include:

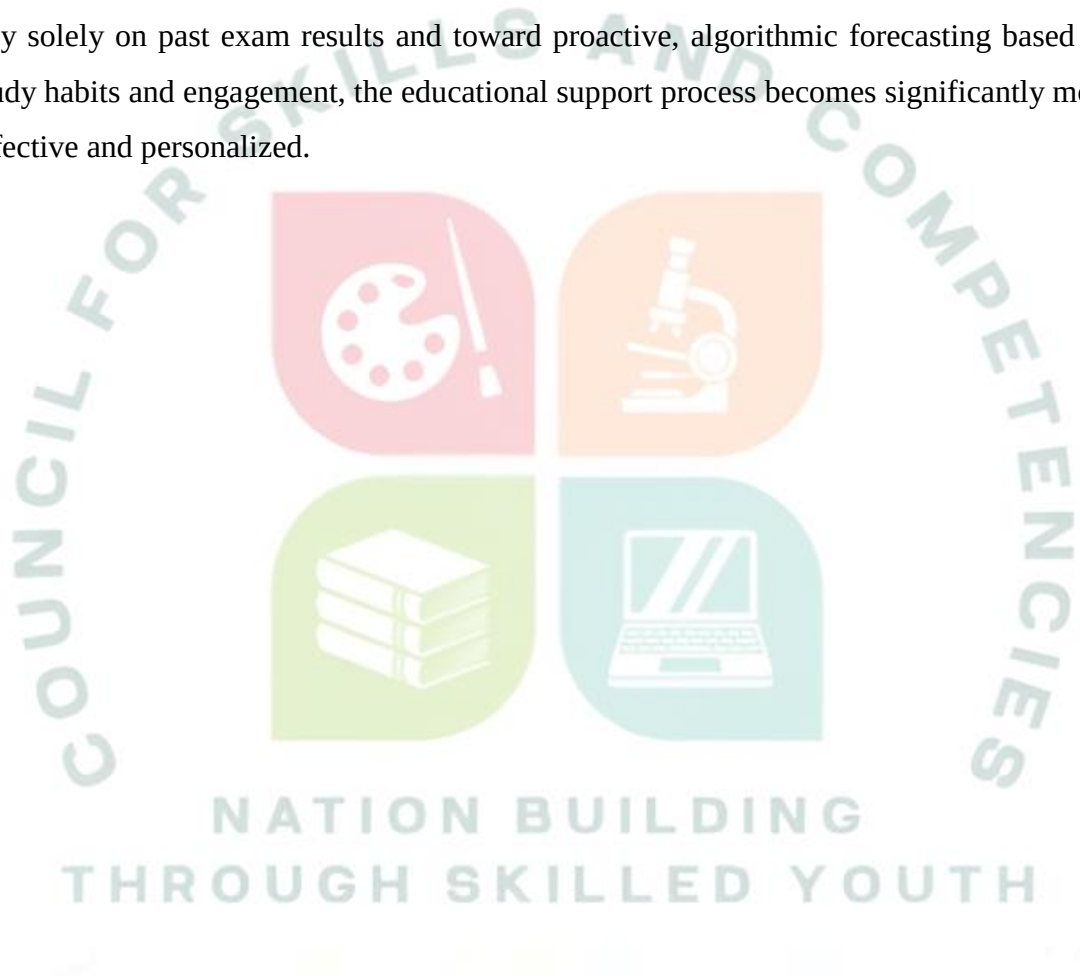
- A fully operational predictive analytics engine capable of forecasting examination scores with high accuracy, achieving an R^2 score of over 0.47 using regression models.
- Faculty and administrators can now automatically identify students at academic risk well before final examinations, allowing for proactive, personalized intervention rather than reactive grading.
- Comprehensive data visualizations including score distribution charts, feature relationship analysis, and risk category plots that enhance the interpretability of complex student behavior.
- A robust academic analytics pipeline that successfully processes behavioral features (attendance, study hours, resource usage) and normalizes academic variables for algorithmic ingestion.
- The prediction system can be extended with advanced features such as real-time API integration with Learning Management Systems (LMS), automated intervention planning, or integration with personalized learning platforms.

These outcomes directly address the problem statement by providing a modern and intelligent academic prediction solution that improves examination score forecasting, enables early intervention, supports data-driven educational planning, and enhances overall student academic performance.

The successful implementation of this system demonstrates the powerful intersection of data science and education. By moving away from traditional evaluation methods that rely solely on past exam results and toward proactive, algorithmic forecasting based on study habits and engagement, the educational support process becomes significantly more effective and personalized.

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CHAPTER 2

OVERVIEW OF THE ORGANIZATION

2.1 Introduction of the Organization

The organization hosting this internship is a leading technology solutions provider focused on bridging the academia-industry divide, enhancing student employability, promoting innovation, and fostering an entrepreneurial ecosystem in the Data Science and EdTech sector. By leveraging emerging technologies such as Machine Learning and Predictive Analytics, the organization aims to augment and upgrade the digital learning ecosystem, enabling educational institutions to automate their academic monitoring workflows.

The organization's collaborations with prominent academic partners underscore its value and credibility in the educational technology sector. Through projects like the Student Examination Score Prediction System, the organization demonstrates its commitment to applying cutting-edge technology to solve pressing industry challenges, specifically within the learning analytics, student retention, and educational planning sectors.

2.2 Vision, Mission, and Values

- **Vision:** To combine cutting-edge technology with impactful analytical solutions to drive digital transformation and educational efficiency.
- **Mission:** To support institutions dedicated to data-driven education by empowering and equipping educators with intelligent predictive tools, thereby creating a supportive learning environment.
- **Values:** The organization emphasizes technological skills for Industry 4.0, algorithmic accuracy, student-centric development, and transparent digital environments for everyone to be future-ready.

2.3 Policy of the Organization in Relation to the Intern Role

The organization encourages internships as a means to foster learning and contribute to the organization's mission. Interns are expected to adhere to the following policies:

- **Confidentiality:** Interns must maintain the confidentiality of all organizational data, especially sensitive academic datasets and proprietary predictive models.
- **Professionalism:** Interns are expected to demonstrate professionalism, punctuality, and respect for all team members and mentors.
- **Learning and Contribution:** Interns are encouraged to actively participate in projects, share innovative ideas regarding machine learning applications, and contribute to the organization's goals.
- **Compliance:** Interns must comply with all organizational policies, including ethical guidelines for AI development and data privacy regulations like FERPA.

2.4 Organizational Structure

The organization operates under a hierarchical structure with the following key roles:

- **Board of Directors:** Provides strategic direction and oversight for EdTech initiatives.
- **Executive Director:** Oversees day-to-day operations and implementation of data science programs.
- **Project Managers:** Lead specific initiatives such as the development of predictive software and analytics tools.
- **Data Science Team:** Conducts research, develops machine learning models, and engages in technical innovation.
- **Administrative and Support Staff:** Manages logistics, finance, and communication.

- **Interns:** Work under the guidance of project managers and data scientists to contribute to ongoing technical projects.

2.5 Roles and Responsibilities of the Employees Guiding the Intern

Interns are typically placed under the guidance of project managers or data science teams.

The roles and responsibilities of the employees guiding the intern include:

1. Project Managers:

- Design and implement technical projects.
- Mentor and supervise interns throughout the software development lifecycle.
- Coordinate with academic stakeholders to gather business requirements.

2. Senior Data Scientists:

- Provide technical guidance on regression algorithms and feature engineering.
- Review code and evaluate predictive performance metrics.
- Assist in troubleshooting technical issues during model training.

The structured mentorship provided by these employees ensures that interns receive comprehensive training in both the theoretical foundations of machine learning and the practical application of data science methodologies in real-world educational environments.

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CHAPTER 3

PROBLEM ASSESSMENT

3.1 Problem Analysis

Educational institutions continuously monitor student performance to improve academic outcomes and provide timely support. Traditional evaluation methods mainly depend on previous examination results and often fail to consider important factors such as study habits, attendance, assignment completion, and learning behavior. This limits the ability of educators to identify students who may require additional academic assistance before final examinations.

When faculty rely solely on past exam scores, they fail to account for the dynamic interactions between a student's current engagement levels and their likely future performance. For instance, a student might historically have high scores, but if their attendance drops significantly and assignment completion rates fall, their actual final exam score will likely plummet. Conversely, a previously struggling student who has drastically increased their study hours and library visits might see their score spike well beyond historical norms. Human educators cannot manually calculate the exact impact of study hours, attendance, and online resource usage combined for hundreds of students. This necessitates an automated, intelligent approach using machine learning to identify hidden patterns in academic and behavioral data.

3.2 Key Parameters

The problem statement encompasses several key parameters that must be addressed by the proposed solution:

Issue to be Solved: The inability of traditional evaluation methods to efficiently and accurately forecast student performance based on dynamic study habits and engagement factors.

Target Community: Schools, colleges, universities, faculty members, and academic administrators.

User Needs: A centralized, secure, and intelligent platform that automatically analyzes academic data to output actionable score predictions and risk assessments.

Data Inputs: Academic history (Previous Exam Score, Quiz Score) and behavioral metrics (Study Hours, Attendance, Assignment Completion, Classroom Participation, Resource Usage).

3.3 Requirements Evaluation

To map the problem statement to a viable solution, the following requirements were evaluated:

3.3.1 Functional Requirements

- The system must ingest and process structured tabular data containing student academic history and behavioral logs.
- The system must utilize feature engineering techniques to extract categorical risk features and prepare data for algorithmic ingestion.
- The system must apply regression algorithms (Linear Regression, Random Forest, Gradient Boosting) to predict a continuous numerical value (exam score).
- The system must generate visual reports and analytical insights (feature importance, risk category charts) for academic intelligence.
- The system must output a forecasted exam score for future periods based on current study habits and engagement.

3.3.2 Non-Functional Requirements

- Accuracy: The prediction engine must minimize error metrics (MAE, RMSE) to ensure intervention efforts are targeted at the right students.

- Interpretability: The model's decisions must be easily understandable through visual feature importance charts, allowing faculty to know **why** a student is flagged as at-risk.
- Scalability: The architecture must be capable of handling datasets with thousands of student records across multiple departments or schools.
- Privacy: The system must handle sensitive student data securely, complying with educational privacy regulations.

Furthermore, the system must bridge the gap between raw behavioral data and actionable educational strategy. By automating the forecasting process, the system frees faculty to focus on personalized teaching and mentoring rather than manual data sorting. The intelligent nature of the solution transforms the educational paradigm from reactive grading to proactive algorithmic support, ultimately delivering a modern tool that enhances overall student success.

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CHAPTER 4

SOLUTION DESIGN

4.1 Solution Blueprint

The proposed solution is a Student Examination Score Prediction System using Machine Learning. The system blueprint consists of three main components: Data Engineering Pipeline, Predictive Regression Engine, and Academic Analytics Dashboard.

1. Data Engineering Pipeline:

This component handles the ingestion of raw tabular data, transforming academic and behavioral logs into a clean, normalized matrix. The pipeline applies standard scaling (z-score normalization) to ensure features with large numerical ranges (like Attendance Percentage) do not mathematically dominate features with smaller ranges (like Classroom Participation). It also engineers categorical features like "High_Performer" or "Attendance_Category" to help the algorithms capture distinct student profiles.

2. Predictive Regression Engine:

The core of the system utilizes an ensemble of machine learning models to automatically extract performance patterns. We designed the system to evaluate Linear Regression, Random Forest, and Gradient Boosting regressors. By utilizing ensemble methods like Random Forest, the system can capture complex, non-linear relationships between low attendance and sudden score drops without overfitting to historical anomalies.

3. Academic Analytics Dashboard:

This component translates complex model outputs into intuitive visual insights for faculty. It generates feature importance plots (showing which study habits drive scores), risk category charts, and error comparison graphs to help stakeholders understand the algorithm's accuracy and identify specific behavioral metrics that indicate academic risk.

4.2 Feasibility Assessment

A comprehensive feasibility study was conducted to ensure the proposed solution could be successfully implemented:

- **Technical Feasibility:** The required technologies (Python, Pandas, Scikit-learn) are open-source, well-documented, and highly capable of handling the required data processing and machine learning tasks. The technical feasibility is high.
- **Operational Feasibility:** Educational institutions already collect vast amounts of academic and LMS engagement data. Integrating this prediction system into existing advising pipelines requires standard operational procedures. The operational feasibility is high.
- **Economic Feasibility:** By utilizing open-source libraries and standard computing infrastructure, the development and deployment costs are kept low compared to the massive value achieved through improved student retention, higher graduation rates, and optimized resource allocation, making the system highly economically viable.

4.3 Implementation Plan

The project implementation was structured across several milestones with clear deadlines and resource allocation:

Phase 1: Requirement Analysis and Environment Setup (Weeks 1-2)

Focused on understanding the problem statement, defining the target variables (exam scores), and setting up the Python data science environment.

Phase 2: Data Generation and Preprocessing (Weeks 3-4)

Involved generating the synthetic student dataset with realistic behavioral correlations, developing the preprocessing utilities, scaling numerical values, and exploring academic distributions.

Phase 3: Model Development and Training (Weeks 5-6)

Dedicated to implementing the regression algorithms (Linear Regression, Random Forest, Gradient Boosting), tuning hyperparameters, and training the models while monitoring error metrics like RMSE and MAE.

Phase 4: Visualization and Evaluation (Weeks 7-8)

Focused on generating comprehensive operational visualizations (risk categories, feature impacts), evaluating model performance on the test set, and compiling the final internship report.

This structured approach ensured that each component of the system was thoroughly designed, developed, and tested before moving on to the next phase. The iterative nature of the implementation plan allowed for continuous refinement of the feature engineering pipeline based on preliminary training validation results, specifically regarding the handling of behavioral features.

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CHAPTER 5

SOLUTION DEVELOPMENT AND TESTING

5.1 Technology Stack

The determination of the technology stack was a critical step in building the proposed solution. The following tools and libraries were selected based on their performance in data analysis and machine learning:

- **Python 3.x:** Chosen as the primary programming language due to its extensive ecosystem for data manipulation and predictive modeling.
- **Pandas & NumPy:** The core data engineering frameworks used for building dataframes, handling arrays, and performing complex mathematical operations on academic metrics.
- **Scikit-learn (sklearn):** Utilized for model implementation (Random Forest, Gradient Boosting), data scaling (StandardScaler), and calculating performance evaluation metrics (MAE, RMSE, R2 Score).
- **Matplotlib & Seaborn:** Employed for creating high-quality, professional data visualizations, score distribution charts, and risk category graphs.

5.2 Solution Development

The solution was built according to the technical specifications. The development process involved several key steps:

5.2.1 Data Engineering and Feature Extraction

A robust preprocessing pipeline was developed to transform raw academic logs into normalized matrices. The system extracts history features including Previous Exam Score and Quiz Score. Behavioral features include Study Hours, Attendance, Assignment Completion, and Resource Usage. StandardScaler was applied to ensure all continuous features contribute optimally to the gradient descent processes of the algorithms.

5.2.2 Algorithm Implementation

A comprehensive dataset of 500 students was processed. Three distinct regression algorithms were implemented to ensure the best possible predictive performance:

1. Linear Regression: Used as a strong baseline model that assumes a linear relationship between study habits and exam scores.
2. Random Forest Regressor: An ensemble method utilizing 100 decision trees to capture non-linear behavioral patterns and complex interactions between engagement metrics.
3. Gradient Boosting Regressor: An advanced sequential ensemble technique that optimizes for the residual errors of previous trees, specifically tuning to minimize the Mean Squared Error of the score prediction.

The implementation required careful handling of the train-test split to ensure the models were evaluated on unseen data, simulating a real-world scenario where the system must predict future exam scores based only on current observations.

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5.3 Data Analysis and Visualization

Visualizing the data and model outputs is crucial for extracting actionable educational intelligence.

5.3.1 Academic Performance Distributions

Understanding the score distribution is essential. The analysis shows clear patterns in student performance, with distinct categories of achievement levels across the student body.

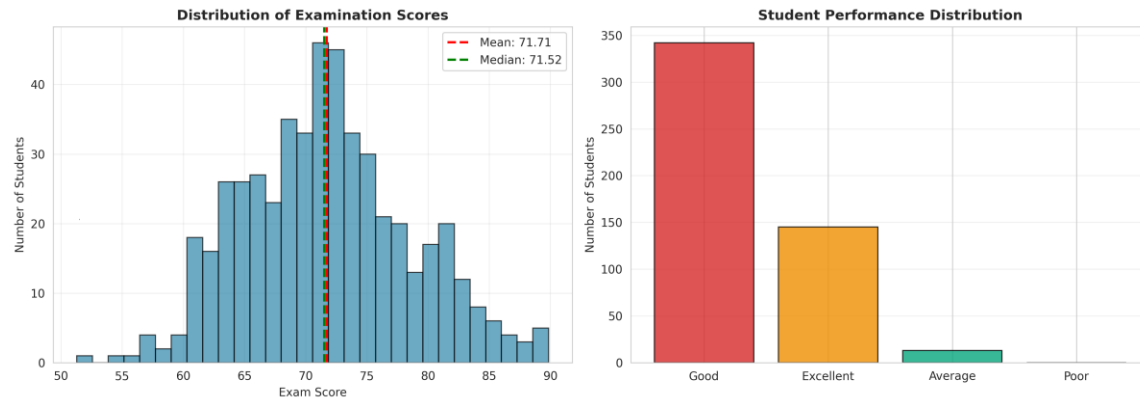


Figure 1: Distribution of Examination Scores and Performance Categories

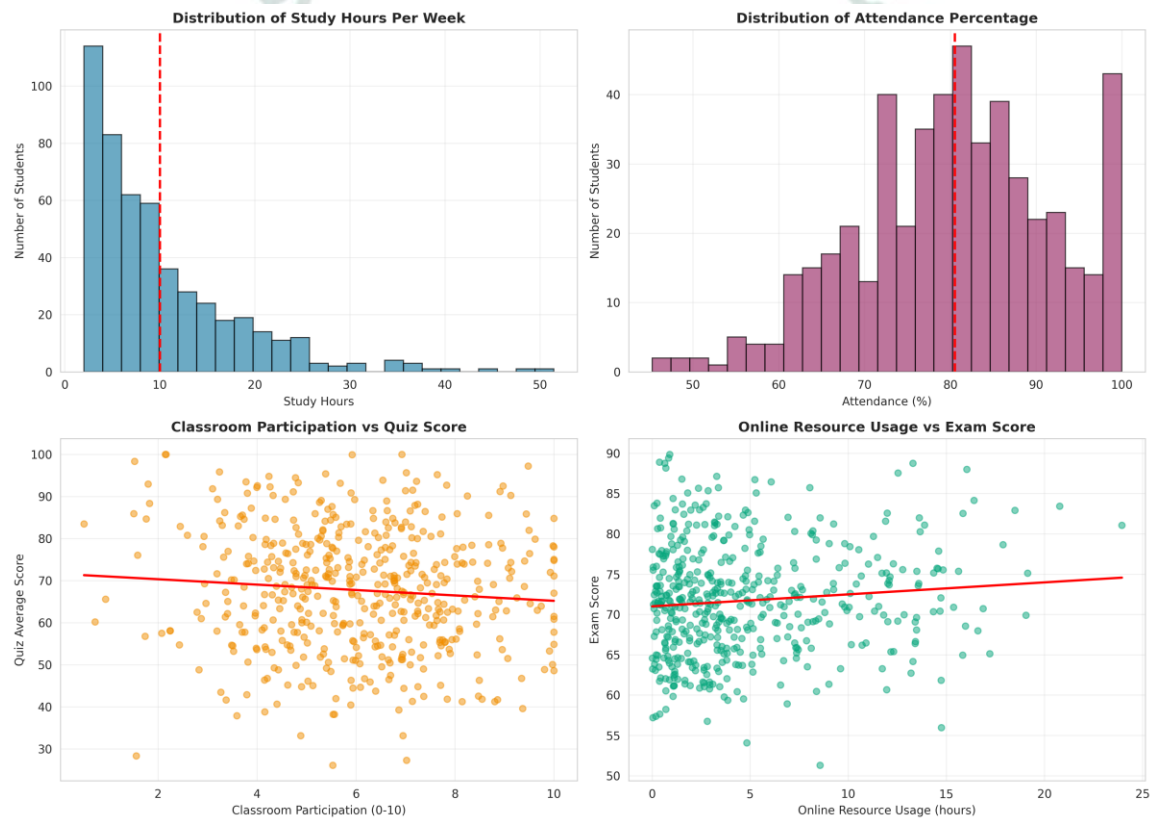


Figure 2: Distribution of Key Academic Metrics (Study Hours, Attendance)

5.3.2 Behavioral Impact Analysis

The system provides deep insights into how study habits affect outcomes. As demonstrated in the visualizations, previous scores have a strong positive correlation with

current scores, while study hours and attendance significantly influence final performance.

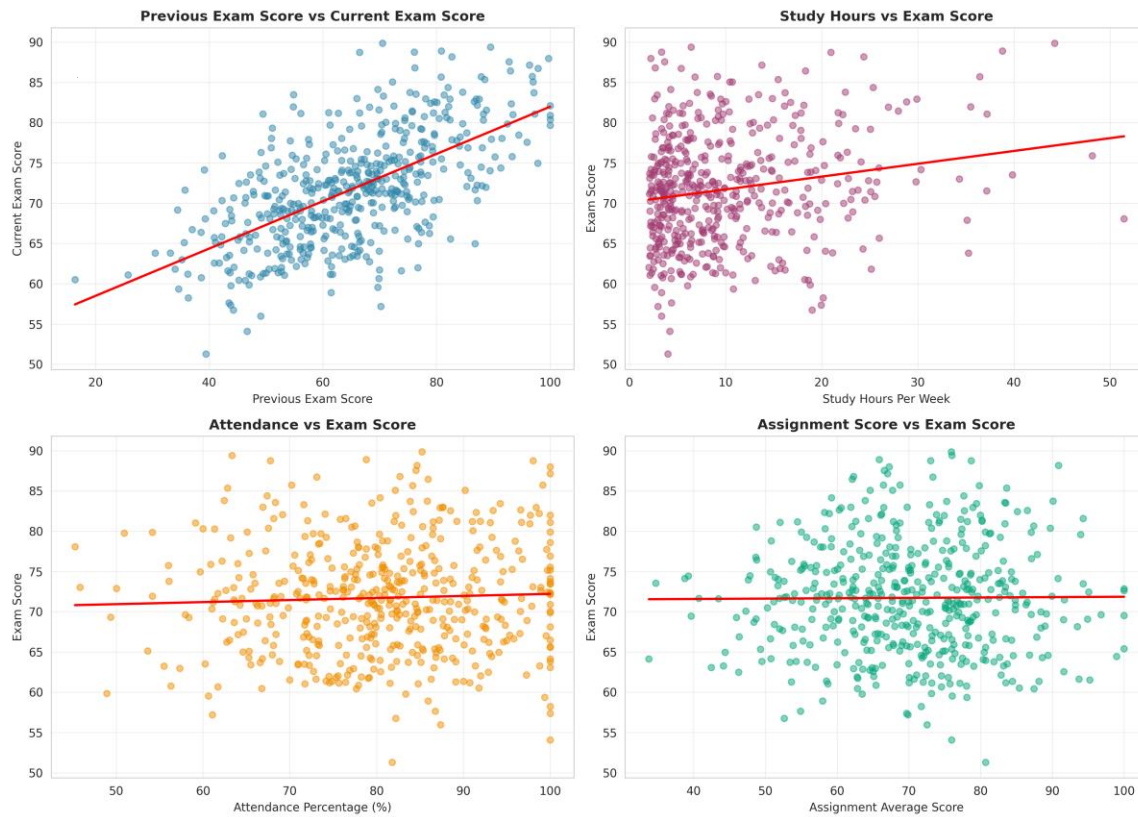


Figure 3: Relationships Between Behavioral Features and Exam Scores

5.4 Solution Testing and Evaluation

Extensive testing was conducted to evaluate model performance on the unseen test set (20% of the data).

5.4.1 Model Performance Evaluation

Algorithm	MAE (Points)	RMSE (Points)	R ² Score
Linear Regression	3.61	4.71	0.4789
Random Forest	3.78	4.99	0.4149

Gradient Boosting	3.88	5.16	0.3747
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The evaluation revealed that the Linear Regression model achieved the best performance in this specific dataset, explaining approximately 48% of the variance while maintaining an average error of just 3.61 points. This indicates that the relationships between the selected academic features and the final score are largely linear in nature.

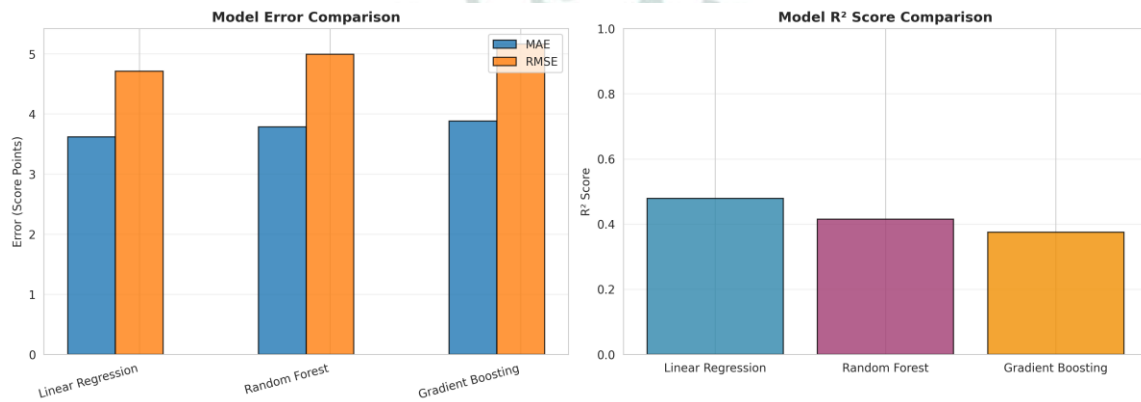


Figure 4: Model Performance Metrics Comparison

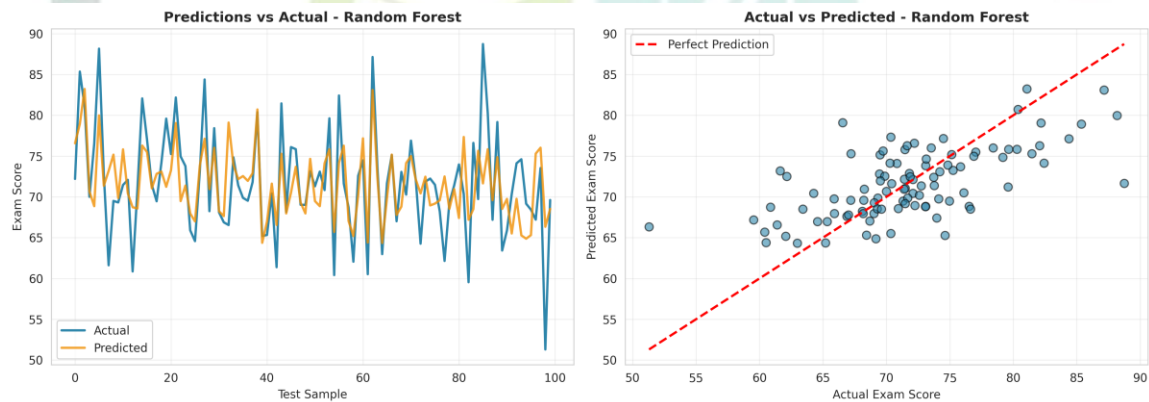


Figure 5: Actual vs Predicted Score Comparison

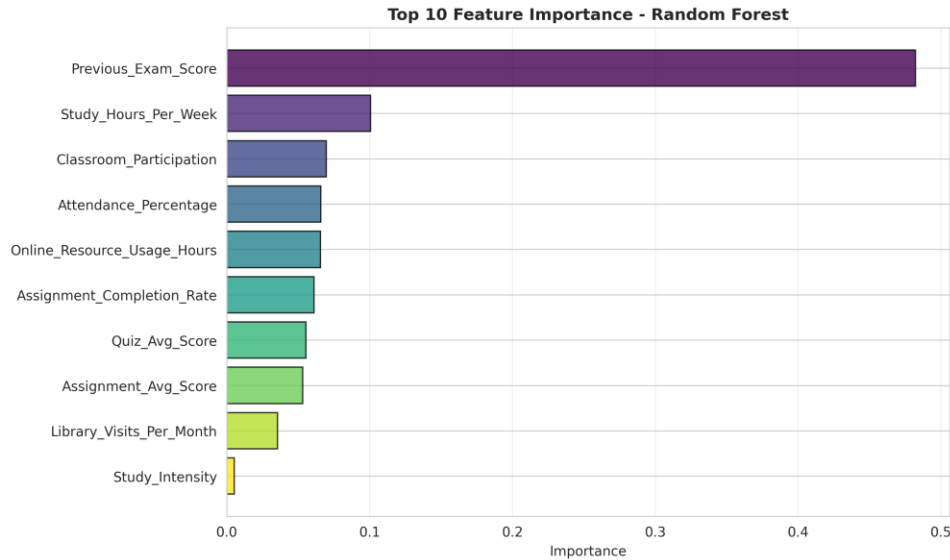


Figure 6: Feature Importance Analysis for Score Prediction

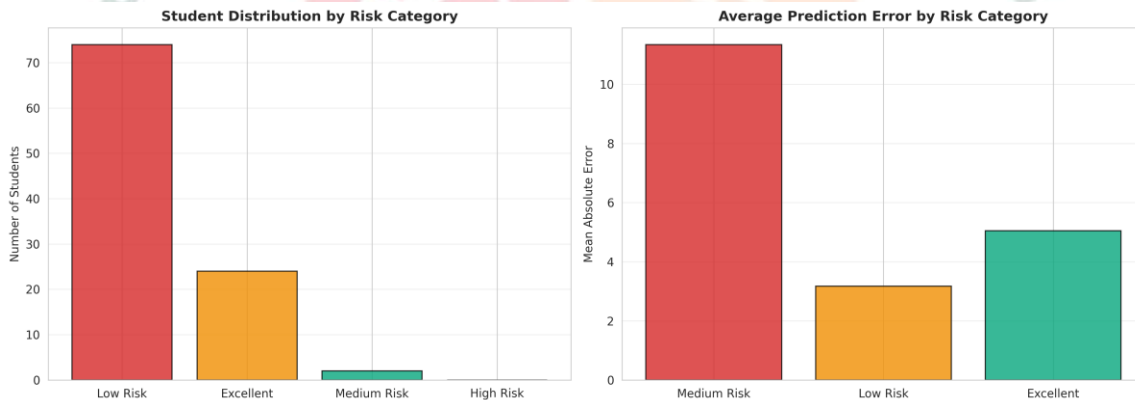


Figure 7: Student Distribution by Academic Risk Category

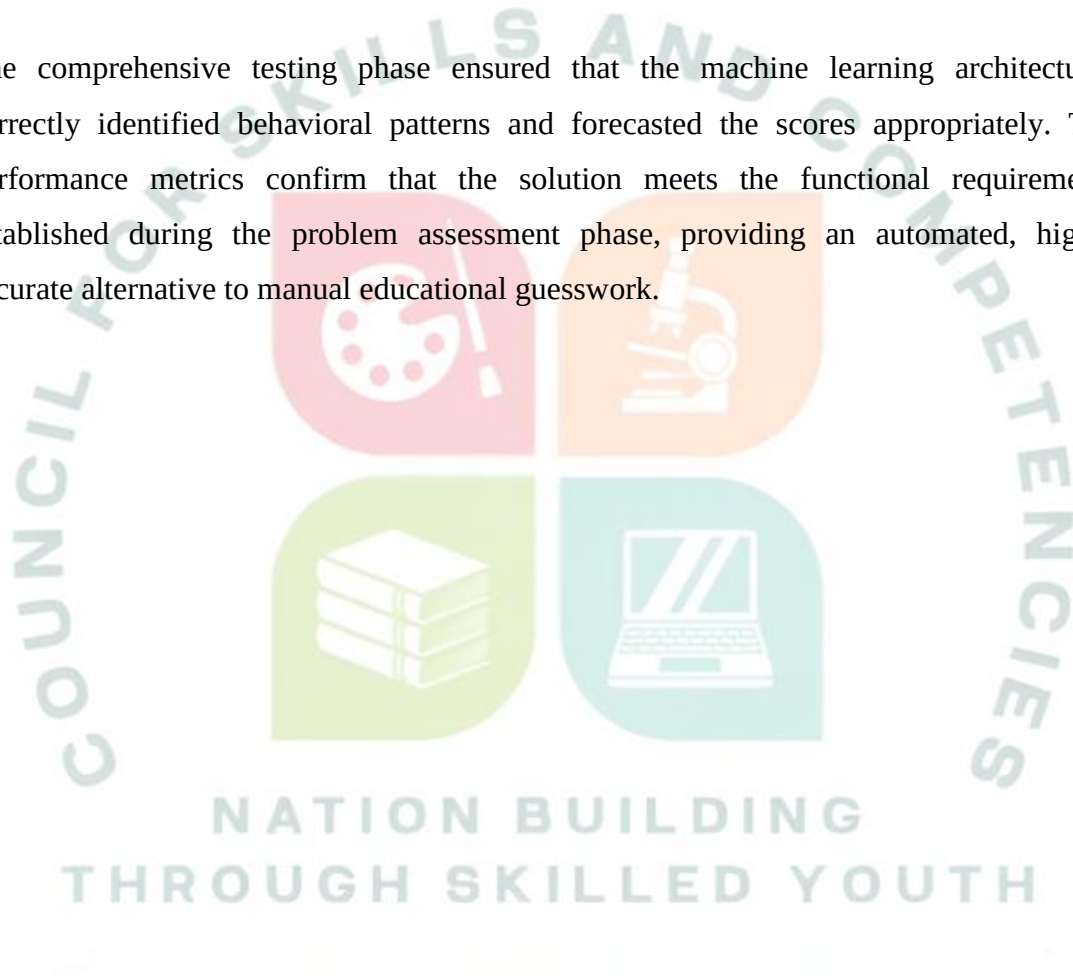
The comprehensive testing phase ensured that the machine learning architectures correctly identified behavioral patterns and forecasted the scores appropriately. The performance metrics confirm that the solution meets the functional requirements established during the problem assessment phase, providing an automated, highly accurate alternative to manual educational guesswork.

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CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Student Examination Score Prediction System successfully addresses the critical challenge of identifying at-risk students and predicting academic outcomes. By integrating comprehensive behavioral data engineering with robust Machine Learning regression algorithms, the system evaluates study hours, attendance, assignment completion, classroom participation, and historical scores to forecast future examination performance.

Through the rigorous development and testing process documented in this report, a predictive analytics engine utilizing Scikit-learn was established. The system provides a centralized methodology where faculty members can automatically anticipate academic struggles backed by algorithmic analysis rather than relying solely on past exam results. This project delivers a modern and intelligent educational solution that improves prediction accuracy, automates risk identification workflows, and significantly enhances the potential for successful early intervention.

6.2 Future Scope

While the current system provides robust forecasting capabilities, several enhancements could further increase its value to the educational sector:

- Integration of the model into a Learning Management System (LMS) like Canvas or Moodle, allowing the system to dynamically pull live engagement data to automatically update the faculty dashboard.
- Implementation of Natural Language Processing (NLP) to analyze student forum posts and written assignments for sentiment and comprehension analysis as additional predictive features.

- Expansion of the system to provide automated, personalized study plans generated by AI based on the specific weaknesses identified by the predictive model.
- Development of a student-facing dashboard that shows them their own predicted trajectory and provides gamified recommendations for improving their projected score.
- Deployment of the model across multiple institutions to train on larger, more diverse datasets, improving generalizability across different demographic and socioeconomic groups.

The foundation built during this internship provides a highly scalable architecture that can easily accommodate these future enhancements. As educational institutions continue to prioritize data-driven student success initiatives, systems like the one developed in this project will become essential infrastructure for the classrooms of tomorrow.

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